

Worksheet · Section 2.1

Binary Numbers



The Big Picture

Everything in a computer is bits. Practice converting between binary and decimal, figuring out how many bits you need, and spotting overflow and roundoff.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|-------------------------------|--|
| ____ 1. Bit | A. Measuring analog data at discrete intervals to store it digitally |
| ____ 2. Byte | B. The base-2 number system |
| ____ 3. Binary | C. A binary digit: 0 or 1 |
| ____ 4. Decimal | D. The base-10 number system |
| ____ 5. Place value | E. What determines whether bits represent a number, a letter, a color, etc. |
| ____ 6. Analog data | F. A real number is stored as an approximation because bits are limited |
| ____ 7. Sampling | G. Data whose values change smoothly over time |
| ____ 8. Overflow error | H. A group of 8 bits |
| ____ 9. Roundoff error | I. A value is too large for the number of bits available |
| ____ 10. Context | J. What a digit is worth based on its position (powers of 2 in binary) |

Part 2 · Fill It In

Use each word from the word bank once.

double**sampling****bits****16****roundoff****8****2****256**

1. Binary is base ____.
2. One byte is ____ bits.
3. Four bits can represent ____ different values.
4. Eight bits can represent ____ different values.
5. Adding one more bit will ____ the number of values.
6. All data in a computer is ultimately stored as ____.
7. Recording sound by measuring it 44,100 times per second is ____.
8. When $0.1 + 0.2$ doesn't equal exactly 0.3 , that's a ____ error.

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Practice

Part 3 · Convert It

Binary	Decimal
101	_____
1111	_____
10000	_____
101010	_____
_____	9
_____	24
_____	63
_____	100

Use the place values: $64 \cdot 32 \cdot 16 \cdot 8 \cdot 4 \cdot 2 \cdot 1$

Part 4 · How Many Bits?

Unique values needed	Minimum bits
10	_____
26 letters	_____
50	_____
200	_____
1,000	_____

Largest value with 5 bits: _____ · with 7 bits: _____

Rule: n bits hold _____ values; the largest is _____.

Part 5 · Compare & Order

Order from least to greatest: 1010, 0111, 1100, 1001

→ _____

Which is larger, 100000 or 11111? _____

How can you tell a binary number is odd just by looking? _____

Part 6 · Overflow Detective

A 4-bit counter starts at 0 and adds 1 every second.

After 15 seconds it shows _____. At 16 seconds an _____ error happens and it wraps to _____.

A game stores lives in 3 bits. The most lives possible is _____.

Fix the game so it can store up to 100 lives: use at least _____ bits.

**Think:**Each extra bit **DOUBLES** the number of values. To find the bits you need, keep doubling 1, 2, 4, 8, 16... until you reach the number of values.

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What is binary **11001** in decimal?

- A. 11001
- B. 25
- C. 27
- D. 19

★★ On Level

2. What is decimal **50** in binary?

- A. 110011
- B. 110100
- C. 110010
- D. 101010

★★ On Level

3. A system uses 3 bits for each color channel. How many different shades can each channel have?

- A. 3
- B. 8
- C. 6
- D. 9

★★★ Exam Hard

4. A program must give each of the 50 U.S. states a unique binary code using the fewest bits. How many bits are needed?

- A. 5
- B. 7
- C. 50
- D. 6

★★★ Exam Hard

5. Which TWO statements are true? (*Select two.*)

- A. Adding a bit adds one more possible value.
- B. The same bits can represent a number or a character, depending on context.
- C. 8 bits can represent 256 different values.
- D. The largest 8-bit value is 256.

Part 8 · Written Response

Explain why using more bits per sound sample makes a recording sound better, and describe the trade-off.
